

BiomechCoach

Computer Vision Biomechanical Coaching — Mobile App
MVP Specification & Execution Plan

USTA Connect Innovation Challenge 2026 · Powered by Rally AI

1. Product Overview

BiomechCoach is a mobile app (iOS + Android) that analyses a tennis swing from a smartphone video and delivers elite-level biomechanical feedback via Rally, the built-in AI coaching assistant. Upload a clip, receive a skeleton overlay with joint measurements, and get personalized coaching notes with specific drills — all from your phone on court.

The Problem

Elite biomechanical coaching costs \$200/hr and only exists at top academies. Kids in underserved areas never access it. The playing field is tilted before they step on court.

The Solution

MediaPipe pose estimation + custom metrics engine + Rally AI coaching layer — delivered as a native mobile app. Any player with a smartphone gets elite analysis for free, right on court.

Platform

iOS + Android

Mission impact

Strong

Innovation

Highest

Demo risk

Low

USTA data

Central

Timeline fit

Tight

Rally AI uses USTA performance data as the elite baseline — e.g. "Your elbow angle is 18° below the average USTA 4.0+ player at contact." USTA data is central to every metric output, not an afterthought. Judges demo the app by scanning a QR code in Expo Go — no App Store submission required.

2. MVP Features

Player records or uploads a 5–30 sec clip from the phone camera roll. Selects shot type: serve, forehand, backhand, or volley. Auto-detects swing window via motion detection so analysis focuses on the correct frames.

Must have

MediaPipe Pose extracts 33 body landmarks per frame. Coloured dots on every joint connected by lines, overlaid on the video. Problem joints highlight red, good joints orange. Three tabs: Your Form / Ideal Form / Compare overlay side by side.

Must have

Calculates tennis-specific metrics: elbow angle at contact, shoulder rotation degrees, hip-to-shoulder timing, wrist snap speed (optical flow), weight transfer, knee bend depth.

Must have

Player metrics compared to USTA 4.0+ aggregated baselines. Score card colour-coded: orange = on target, gray = slightly off, red = significantly off. Percentile rank shown.

Must have

Claude API (claude-sonnet-4-6) converts metric gaps into plain-English coaching cards: Fix This / Looking Good / Pro Tip. Every piece of feedback references the player's actual numbers. Language calibrated to player's NTRP level.

Must have

Coaches see their player roster with USTA ranking, NTRP rating, biomechanical score, and a spark line trend per player. Simple, defensible, and directly tied to USTA data.

Must have

Weekly session history with biomechanical score trends and metric improvement charts. Shows coaches and players whether form work is translating over time.

High priority

Real-time skeleton overlay via device camera using MediaPipe JS. Build in final week of July — only include in demo if stable on the exact device used for the showcase.

Stretch

One-page PDF summary: skeleton screenshot, metric scores, baseline comparison, drill list. Player shares with coach or parent via phone share sheet.

Nice to have

3. Computer Vision Pipeline

1	2	3	4	5	6	7
Video upload	Frame extract	Swing detect	Pose estimate	Metrics calc	Baseline compare	Rally AI

1

Player selects video from camera roll via Expo ImagePicker. File uploaded to Supabase Storage via the FastAPI backend. Async processing job triggered immediately.

2

Reads video at 30fps. Motion detection pass (frame differencing) identifies the swing window — typically 1–3 seconds of peak motion. Only those frames proceed.

3

Google MediaPipe Pose extracts 33 3D body landmarks per frame. Runs on CPU — no GPU required. Free and open source. No custom model training needed.

4

Joint angles (dot product of adjacent vectors), wrist velocity (optical flow), centre-of-mass shift (hip trajectory), shoulder-to-hip rotation timing (frame delta).

5

Player metrics compared against aggregate profiles of high-rated USTA players. Scores expressed as percentile. This is where USTA data becomes central to every output.

6

Metric gaps + shot type + NTRP level sent to Claude API with the Rally system prompt. Returns structured JSON: Fix This / Looking Good / Pro Tip cards with actual player numbers embedded in every response.

Risk	Mitigation
	MediaPipe struggles in poor lighting. Onboarding screen with camera setup guidelines. Always use pre-recorded controlled videos for the showcase demo.
	CV pipeline runs on the FastAPI backend server, not the device — so phone specs don't affect analysis speed. Async processing with progress indicator in the app.

Get a USTA-certified pro to validate Rally's feedback on 10 sample swings before submission. Document the validation — your credibility shield with judges.
If USTA data lacks biomechanical detail, use ITF/ATP public stats as proxy baseline. Decide and implement this contingency before July 11.
Test the QR code flow on the exact judge devices 24 hours before the showcase. Have a screen-recorded backup video ready as a fallback.

4. Technology Stack

Mobile Frontend

Framework	React Native + Expo
Video upload	Expo ImagePicker + Expo Camera
Skeleton overlay	React Native Skia (canvas drawing on video)
Charts + progress	Victory Native (metrics, spark lines, progress bars)
Navigation	Expo Router (file-based routing)
Distribution	Expo Go (QR code — no App Store submission needed)

Backend + CV

API layer	FastAPI (Python) — native for CV workloads
Pose estimation	MediaPipe Pose — Google, open source, CPU-only
Video processing	OpenCV — frame extraction, optical flow, motion detection
Metrics layer	NumPy + custom Python — joint angle calculations
Rally AI	Claude API — claude-sonnet-4-6
File storage	Supabase Storage — video uploads
Database	Supabase PostgreSQL — players + sessions tables
Backend hosting	Railway or Render — free tier Python servers

Data Sources

Primary baseline	USTA player performance data (released July 11)
Fallback baseline	ITF / ATP public match stats for elite benchmarks
Reference swings	Public domain pro footage — YouTube Creative Commons
Drill library	Original content — no third-party IP issues

Repository + License

Version control	GitHub — private repo until after August 9 submission
License	MIT License
.gitignore	Python template + video files, USTA data, .env, node_modules
Demo delivery	Expo Go QR code — judges scan and run instantly

5. Rally AI — System Prompt

Rally is the AI coaching personality inside BiomechCoach, powered by Claude API (claude-sonnet-4-6). Every piece of feedback references actual player metric numbers and is calibrated to the player's NTRP level. Output is structured JSON mapped directly to the Fix This / Looking Good / Pro Tip card components in the app.

You are Rally, an expert AI tennis biomechanics coach built into the BiomechCoach app. Analyse the biomechanical metrics from the player's swing and return structured coaching feedback. Rules: never give generic advice — every point must reference the player's actual metric numbers. Keep language plain and calibrated to the player's NTRP level. Drills must be actionable with no equipment unless specified. Never invent baseline numbers — only use the USTA baseline values provided in the input. Output only valid JSON — no markdown, no preamble.

Input (sent from FastAPI)	
shot_type	serve / forehand / backhand / volley
ntrp_level	player's NTRP rating e.g. 3.5
metrics	elbow_angle, shoulder_rotation, wrist_snap, weight_transfer, hip_timing, knee_bend
each metric	{ value, baseline, unit } — all from USTA data or validated proxy

Output (JSON returned to app)	
fix_this	array of { title, body, drills[] } — 1 to 3 issues max
looking_good	array of { title, body } — 1 to 2 strengths
pro_tip	{ title, body } — one advanced kinetic chain insight

6. Execution Timeline

NOW → JULY 3

File the USTA entry form. Set up GitHub repo (private, Python .gitignore, MIT license). Scaffold FastAPI backend and Expo React Native frontend. Start MediaPipe + OpenCV pipeline on test videos today — don't wait for USTA data.

JULY 3 – 10

Get pose extraction working end-to-end: upload video, extract landmarks, return skeleton JSON to the app. Build the three-tab skeleton overlay screen (Your Form / Ideal Form / Compare) using mocked landmark data. Wire Rally AI prompt to Claude API.

JULY 11 – 14

On data release day, immediately assess available performance data. Extract high-rated player stats to build the USTA elite baseline. If biomechanical data is absent, activate the ITF/ATP proxy baseline — decide and implement this contingency now.

JULY 15 – 21

Run 10–15 test swings through the full pipeline. Have a USTA-certified pro review Rally's feedback for accuracy. Document the validation. Build the coach roster view with USTA ranking, NTRP rating, biomechanical score, and spark line trend.

JULY 22 – 28

Polish all five screens. Record 3 clean demo videos (serve, forehand, backhand) under controlled lighting. Attempt live mode implementation — only include in demo if stable on the exact showcase device.

JULY 29 – AUG 4

Record the 3–5 minute demo walkthrough. Write the 5-page overview. Build SBOM (MediaPipe, OpenCV, NumPy, Expo — all MIT/Apache licensed). Document equity narrative.

AUG 5 – 9

Final submission: Expo Go QR code + hosted backend link, written overview, SBOM, demo video, license disclosures. Freeze new features by Aug 5. Last 4 days = polish only.

7. US Open Showcase — Demo Plan (Sept 3, 5 minutes)

"The difference between a player who goes pro and one who doesn't often comes down to a few degrees of elbow angle. Elite academies know this. BiomechCoach puts that knowledge in every player's pocket — on any smartphone, for free."

0:00 – 0:30

The hook

Set the democratisation framing. One in a million kids has access to a biomechanics lab. Pull out your phone. BiomechCoach changes that.

0:30 – 2:00

The skeleton moment — your showstopper

Play pre-recorded serve video in the app. Skeleton dots and connecting lines appear over the body frame by frame. Problem joints light up red. Pause at contact — elbow 94° in red vs 112° elite baseline annotated on screen. Let it breathe 5 seconds. Judges remember this.

2:00 – 3:15

The Compare tab + Rally feedback

Switch to Compare overlay — orange skeleton vs faded green ideal form simultaneously. Scroll to Rally's coaching cards. Read out a Fix This card with the actual degree numbers. Show the drill list. This is where Rally's personality shines.

3:15 – 4:15

Coach roster view

Switch to the coach view. Show the player roster — USTA ranking, NTRP rating, biomechanical score, spark line trend. 'A coach can see in one screen which players are improving and which need attention — all tied directly to their USTA data.'

4:15 – 5:00

The equity close

"A kid at IMG Academy gets this every day. A kid in rural Tennessee gets none — not because they're less talented, but because their family can't afford a biomechanics coach. BiomechCoach is that coach. On any smartphone. For free. That's how you reach 35 million."

1. Always demo using pre-recorded videos — never rely on a live camera at the showcase.

2. Have 3 backup videos ready on the device: serve, forehand, and backhand.

3. Test the Expo Go QR code on the exact judge device 24 hours before the showcase.

4. Have a screen-recorded video fallback ready if Expo Go or the backend fails.

5. Backend must be deployed and live on Railway/Render — no localhost on demo day.

